

DAEN 427 Homework 2 - Alexander Abuabara

Due date: Aug 28, 2025

Rubric: any reasonable analysis using Bayesian approach may receive 10 points.

1. Why is it that we identify ourselves at the very beginning of a phone conversation, even to people we already know, but we do not do this when we meet in person? Express your answer within the framework of Bayesian perceptual inference.

Answer: A possible explanation is that identifying someone by their voice (like during a phone call) is less reliable than recognizing them visually in a face-to-face conversation. To reduce this uncertainty, people often introduce themselves when speaking on the phone.

In Bayesian terms our brain computes:

$$P(\text{Identity}|\text{Sensory Input}) \propto P(\text{Sensory Input}|\text{Identity}) \cdot P(\text{Identity})$$

Where:

- Posterior = The probability of a person's identity given the sensory input.
- Likelihood = The probability of the sensory input given that identity.
- Prior = The brain's expectation about who it might encounter.

On the phone, sensory input is impoverished, you only have voice without visual or contextual cues (face, gestures, location, body language). The likelihood function $P(\text{Sensory Input}|\text{Identity})$ is broad and noisy because many voices sound similar. The prior may also be weak unless the call was expected. Therefore, the posterior probability $P(\text{Identity}|\text{Sensory Input})$ is highly uncertain. So, explicitly stating your identity helps the other person "jump" to the correct inference and resolve ambiguity. You're essentially supplying the missing input to update their Bayesian model more confidently.

In person, the sensory input is rich:

- Voice,
- Visual appearance (face, clothes),
- Location/context (school, office, etc.).

This makes the likelihood function highly informative, narrowing down identity quickly. The prior may also be strong if the setting suggests who you're likely to see. The posterior inference of identity is quick and confident, so there's no need to explicitly state who you are—your identity is automatically inferred from sensory evidence.

So, we identify ourselves on the phone because the perceptual inference problem is underspecified and we help solve it by supplying a key piece of evidence.

2. When a conversation partner speaks softly, or when a conversation occurs in the presence of significant ambient noise, we sometimes cup our ears and/or look carefully at the speaker's lips. Why, in Bayesian perceptual terms, do we do this?

Answer: In Bayesian perceptual terms, cupping our ears or looking at the speaker's lips in noisy or quiet situations is a strategy to improve the quality of sensory evidence, thereby enhancing the accuracy of perceptual inference under uncertainty.

Bayesian perceptual interpretation:

- Cupping the ears: enhancing the likelihood (sensory evidence). This action increases the signal-to-noise ratio by capturing more sound and focusing it toward the eardrum. In Bayesian terms, this improves the reliability of the sensory evidence, narrowing the likelihood distribution. A narrower likelihood means your brain can update its beliefs (posterior) more confidently based on what it hears.
- Lip-reading / watching the speaker's mouth: multisensory integration to reduce uncertainty. Vision provides redundant cues about speech (e.g., lip shapes correlate with phonemes). Your brain integrates visual and auditory signals in a statistically optimal way (e.g., via Bayesian cue integration). If the auditory cue is noisy but the visual cue is more reliable, the brain weights the visual cue more heavily in forming the posterior. This reduces uncertainty in the posterior belief about what was said, even if the sound itself is unclear.

3. Design a mini experiment to investigate whether listeners' accuracy, reaction times, and confidence differ when interpreting speech from speakers of different genders and accents. Provide:

- A Bayesian model that frames the research question (in terms of priors, likelihoods, and posteriors).
- A short paragraph discussing potential biases (that may be implicit and/or represented in the priors).
- A short paragraph discussing ethical implications of perceptual bias in speech interpretation.
- A brief methods description, including details about participants, experimental conditions, and materials.
- Design:
 - A hand-coded sample data table representing how individual participant data might look,
 - A summary results table showing averages across observations.

Answer: Example of research question: *How do speaker gender and accent influence listener accuracy, reaction times, and confidence?*

Let the observed outcomes be:

- Y_{1i} = Accuracy (binary; correct/incorrect)
- Y_{2i} = Reaction time (continuous)
- Y_{3i} = Confidence (continuous)

Predictors:

- $G_i \in \{0, 1\}$: Gender of speaker (0 = Male, 1 = Female)
- $A_i \in \{0, 1\}$: Accent of speaker (0 = Native, 1 = Non-native)

Priors: For all outcomes $j = 1, 2, 3$:

$$\begin{aligned}\alpha_j &\sim \mathcal{N}(0, 5) \\ \beta_{G,j} &\sim \mathcal{N}(0, 2) \\ \beta_{A,j} &\sim \mathcal{N}(0, 2) \\ \beta_{GA,j} &\sim \mathcal{N}(0, 2) \\ \sigma_j &\sim \text{Half-Cauchy}(0, 2.5) \quad (\text{for continuous outcomes})\end{aligned}$$

Likelihoods: Accuracy:

$$\begin{aligned}Y_{1i} &\sim \text{Bernoulli}(p_i) \\ \text{logit}(p_i) &= \alpha_1 + \beta_{G,1}G_i + \beta_{A,1}A_i + \beta_{GA,1}G_iA_i\end{aligned}$$

Reaction time:

$$\begin{aligned}Y_{2i} &\sim \mathcal{N}(\mu_{2i}, \sigma_2) \\ \mu_{2i} &= \alpha_2 + \beta_{G,2}G_i + \beta_{A,2}A_i + \beta_{GA,2}G_iA_i\end{aligned}$$

Confidence:

$$\begin{aligned}Y_{3i} &\sim \mathcal{N}(\mu_{3i}, \sigma_3) \\ \mu_{3i} &= \alpha_3 + \beta_{G,3}G_i + \beta_{A,3}A_i + \beta_{GA,3}G_iA_i\end{aligned}$$

Posterior: Let Θ denote the full set of parameters:

$$\Theta = \{\alpha_j, \beta_{G,j}, \beta_{A,j}, \beta_{GA,j}, \sigma_j\}_{j=1}^3$$

Then the joint posterior is:

$$P(\Theta \mid \mathbf{Y}, \mathbf{G}, \mathbf{A}) \propto P(\mathbf{Y} \mid \mathbf{G}, \mathbf{A}, \Theta) \cdot P(\Theta)$$

- **Potential biases:** Even with uninformative priors (e.g., Beta(2,2)), assumptions may embed biases. For example, symmetry between gender and accent groups assumes equal baseline performance, which may not reflect societal or linguistic realities. Hierarchical models can mitigate bias by pooling information across groups while allowing variability.
- **Ethical implications:** Biases in speech interpretation can reveal systemic stereotypes. Such results must be ethically framed to avoid reinforcing negative narratives. Ethical procedures include anonymization, debriefing, and stressing that any observed differences reflect listener perception, not speaker traits.
- **Methods**

Participants: 30 native English-speaking adults, gender-balanced.

Materials: 8 speakers reading controlled sentences.
(2 each: male/female x standard/regional accent)

Procedure: Each participant listens to 40 randomized sentences and:

 - Transcribes the sentence (accuracy recorded)
 - Rates confidence (scale 1–10)
 - Reaction time measured from clip end to response start

Sample Participant Data

Participant	Speaker Gender	Speaker Accent	Accuracy	RT (ms)	Confidence
P01	Female	Standard	1	750	8
P01	Male	Regional	0	1200	5
P01	Female	Regional	1	950	6
P01	Male	Standard	1	800	7
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Sample Summary Results

Speaker Gender	Speaker Accent	Mean Accuracy	Mean RT (ms)	Mean Confidence
Male	Standard	0.89	810	7.2
Male	Regional	0.72	1020	6.0
Female	Standard	0.91	790	7.5
Female	Regional	0.75	980	6.3
...	...	...	...	...